

COOLANGATTA, Australia

WMO: 945920

Lat: 28.168S

Lon: 153.505E

Elev: 5

StdP: 101.27

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	7.8	-0.6	3.6	16.6	1.5	4.2	14.9	9.4	18.2	8.5	18.5	1.8	230	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.3	29.7	23.3	28.9	23.3	28.1	22.7	25.2	27.8	24.6	27.2	24.0	26.7	6.8	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.3	26.8	23.7	18.6	26.4	23.0	17.8	25.9	77.3	27.9	74.5	27.2	72.1	26.9	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	9.0	8.2	DB	3.2	33.5	1.2	2.0	2.3	34.9	1.6	36.1	1.0	37.3	0.1	38.7	
			WB	2.2	26.2	1.1	0.6	1.4	26.6	0.8	27.0	0.2	27.3	-0.6	27.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.2	24.5	24.4	23.3	21.1	18.3	16.2	15.3	16.0	18.2	20.0	21.9	23.3
	DBStd	3.73	1.51	1.60	1.47	1.59	1.98	2.13	2.05	2.12	2.05	1.97	1.99	1.76
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	302	0	0	0	2	24	67	97	79	27	6	1	0
	CDD10.0	3718	451	402	414	332	258	186	163	184	247	311	356	413
	CDD18.3	977	192	169	156	83	24	3	2	5	23	59	106	155
	CDH23.3	5077	1329	1123	824	270	44	3	4	16	62	156	413	833
	CDH26.7	637	206	187	86	11	0	0	1	1	8	16	36	87
Wind	WSAvg	4.2	4.7	4.6	4.3	4.0	3.5	3.6	3.5	3.8	4.4	4.7	4.8	4.7
Precipitation	PrecAvg	1710	183	205	241	177	166	112	90	58	51	122	131	156
	PrecMax	2810	515	514	741	728	509	634	393	167	231	640	398	462
	PrecMin	962	16	15	23	38	2	6	0	2	0	0	2	22
	PrecStd	522	123	119	195	147	119	132	90	43	49	132	93	108
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	30.5	29.8	27.9	25.8	23.7	23.5	25.3	27.4	28.8	29.8	30.2
		MCWB	24.5	24.4	23.9	21.3	18.7	17.4	16.1	16.2	16.4	18.1	19.9	21.6
	2%	DB	29.2	29.3	28.2	26.7	24.5	22.4	22.1	23.1	24.8	25.9	27.2	28.2
		MCWB	23.8	23.9	22.9	20.7	18.5	17.0	16.0	16.3	17.8	19.9	21.7	22.7
	5%	DB	28.6	28.7	27.6	25.8	23.6	21.5	21.1	21.9	23.6	24.9	26.3	27.5
		MCWB	23.4	23.5	22.5	20.3	18.1	16.4	15.7	16.0	17.7	19.6	21.3	22.4
	10%	DB	27.9	27.9	26.9	25.0	22.8	20.6	20.1	21.0	22.7	24.0	25.5	26.8
		MCWB	22.9	22.8	21.8	19.9	17.7	15.8	14.9	15.5	17.3	19.1	20.8	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.2	26.1	25.2	23.1	22.0	19.6	19.5	19.9	21.1	22.3	24.4	25.0
		MCDB	28.5	28.8	27.7	25.8	23.3	21.6	21.7	22.6	23.0	24.7	26.1	27.7
	2%	WB	25.2	25.3	24.3	22.0	20.5	18.5	18.1	18.5	20.1	21.5	23.2	24.2
		MCDB	27.8	28.1	27.0	25.0	22.9	20.8	20.6	21.3	22.8	23.8	25.7	26.9
	5%	WB	24.5	24.6	23.5	21.3	19.4	17.7	16.9	17.5	19.2	20.8	22.3	23.5
		MCDB	27.2	27.4	26.4	24.5	22.1	20.2	19.6	20.5	22.2	23.4	25.2	26.2
	10%	WB	23.8	23.9	22.8	20.6	18.5	16.9	15.8	16.5	18.5	20.1	21.6	22.9
		MCDB	26.7	26.7	25.8	23.9	21.4	19.5	19.0	19.8	21.6	23.1	24.6	25.6
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.5	6.8	7.5	8.6	8.6	9.9	10.1	9.0	7.6	6.9	6.6
		MCDBR	7.1	7.2	7.6	8.7	9.7	9.6	11.1	11.6	10.3	8.8	7.5	7.5
		MCWBR	3.4	3.4	3.6	4.3	4.9	5.2	6.2	6.4	5.6	5.0	3.8	3.7
	5% WB	MCDBR	5.9	6.1	6.5	7.7	7.3	7.5	8.8	9.2	7.9	7.2	6.4	6.3
		MCWBR	3.4	3.4	3.6	4.3	4.2	4.6	5.7	5.8	5.0	4.6	3.7	3.5
Clear-Sky Solar Irradiance	taub	0.391	0.377	0.360	0.341	0.321	0.314	0.305	0.318	0.360	0.387	0.384	0.397	
	taud	2.500	2.553	2.597	2.627	2.635	2.649	2.657	2.605	2.479	2.431	2.482	2.469	
	Ebn at Noon	951	948	934	905	881	865	890	917	918	927	952	948	
	Edh at Noon	115	106	96	85	76	72	74	85	106	119	116	119	
All-Sky Solar Radiation	RadAvg	6.45	5.87	5.02	4.23	3.52	2.91	3.39	4.28	5.29	5.91	6.45	6.42	
	RadStd	0.51	0.47	0.28	0.21	0.20	0.19	0.27	0.33	0.42	0.51	0.50	0.67	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)		Station Only	N/A	N/A	N/A	+0.45	N/A	-0.45	N/A	N/A	+118	+90	
(47)		Regional (25 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	-64	+182	+110		

Nomenclature: See separate page